Lab Sheet-7

Ashish singh Rawat
Cu24250221

```
import os

import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

from sklearn.preprocessing import StandardScaler

from sklearn.cluster import KMeans

from sklearn.metrics import silhouette_score

# 0. Setup Output Directory

output_dir = "output_images"

os.makedirs(output_dir, exist_ok=True)

# 1. Load Dataset

DATA_URL = "https://raw.githubusercontent.com/sharmaroshan/Clustering-of-Mall-Customers/master/Mall_Customers.csv"

df = pd.read_csv(DATA_URL)

# Clean column headers

df.columns = df.columns.str.strip().str.replace(' ', '_').str.replace('(', '').str.replace(')', '')

# Select features & scale

feature_cols = ['Annual_Income_k$', 'Spending_Score_1-100']

X = df[feature_cols].copy()

scaler = StandardScaler()

X_scaled = scaler.fit_transform(X)

# 2. Elbow & Silhouette Curves

wcss = []

k_values = range(2, 11)

silhouette_scores = []

for k in k_values:
```

```
    km = KMeans(n_clusters=k, init='k-means++', n_init=10, random_state=42)
    labels = km.fit_predict(X_scaled)
    wcss.append(km.inertia_)
    silhouette_scores.append(silhouette_score(X_scaled, labels))
fig, axes = plt.subplots(1, 2, figsize=(14, 5))
# Elbow Curve
axes[0].plot(list(k_values), wcss, marker='o', color='#1f77b4', linewidth=2)
axes[0].set_title('Elbow Method (WCSS vs. K)')
axes[0].set_xlabel('Number of Clusters (K)')
axes[0].set_ylabel('Inertia (WCSS)')
axes[0].grid(True, linestyle='--', alpha=0.6)
# Silhouette Curve
axes[1].plot(list(k_values), silhouette_scores, marker='s', color='#2ca02c', linewidth=2)
axes[1].set_title('Silhouette Score vs. K')
axes[1].set_xlabel('Number of Clusters (K)')
axes[1].set_ylabel('Silhouette Score')
axes[1].grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
# Save Image 1
fig.savefig(os.path.join(output_dir, "elbow_silhouette_analysis.png"), dpi=300,
bbox_inches='tight')
print(f"Saved: {os.path.join(output_dir, 'elbow_silhouette_analysis.png')}")
plt.show()
# 3. Fit Model & Plot Clusters
optimal_k = 5
kmeans = KMeans(n_clusters=optimal_k, init='k-means++', n_init=10, random_state=42)
df['Cluster'] = kmeans.fit_predict(X_scaled)
centroids_original = scaler.inverse_transform(kmeans.cluster_centers_)
```

```
plt.figure(figsize=(9, 6))
palette = ['#e41a1c', '#377eb8', '#4daf4a', '#984ea3', '#ff7f00']
sns.scatterplot(
    data=df,
    x='Annual_Income_k$',
    y='Spending_Score_1-100',
    hue='Cluster',
    palette=palette,
    s=70,
    alpha=0.9
)
# Centroids
plt.scatter(
    centroids_original[:, 0],
    centroids_original[:, 1],
    s=250,
    c='black',
    marker='X',
    edgecolor='white',
    linewidth=1.5,
    label='Centroids'
)
plt.title('Customer Segments (K-Means Clustering, K=5)', fontsize=13)
plt.xlabel('Annual Income (k$)')
plt.ylabel('Spending Score (1-100)')
plt.legend(bbox_to_anchor=(1.02, 1), loc='upper left')
plt.grid(True, linestyle='--', alpha=0.5)
plt.tight_layout()
```

Save Image 2

plt.savefig(os.path.join(output_dir, "customer_segments_clusters.png"), dpi=300, bbox_inches='tight')

print(f"Saved: {os.path.join(output_dir, 'customer_segments_clusters.png')}")

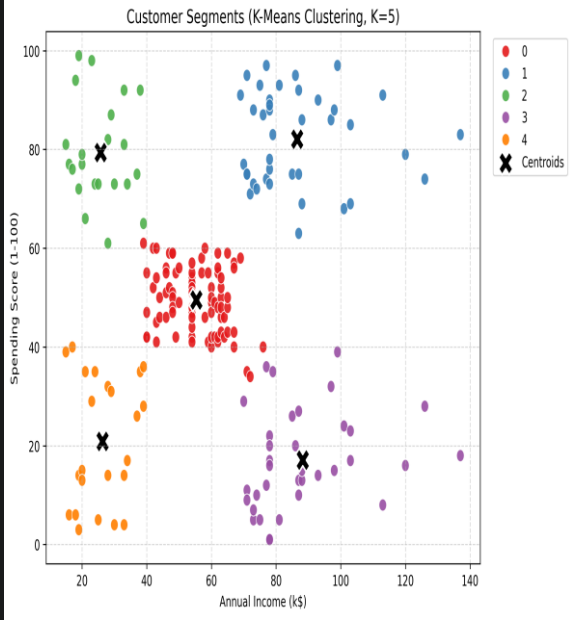
plt.show()

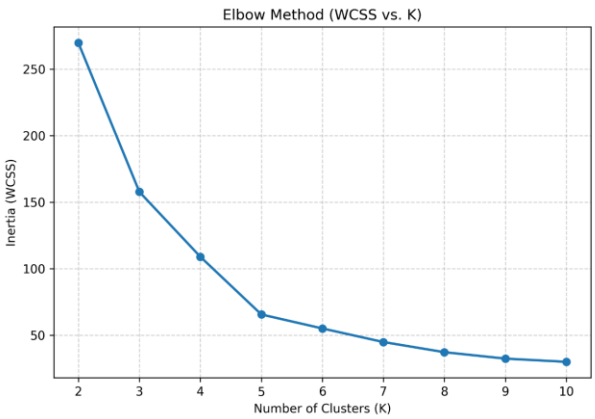
```
Dataset loaded successfully.
Dataset Shape: (200, 5)

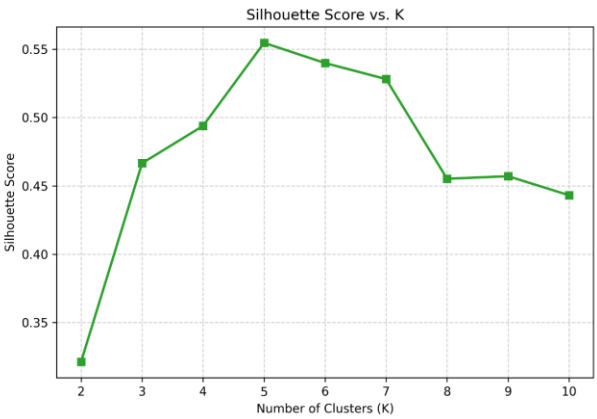
--- First 5 Rows ---
   CustomerID   Genre  Age  Annual Income (k$)  Spending Score (1-100)
0           1    Male   19                  15                      39
1           2    Male   21                  15                      81
2           3  Female   20                  16                       6
3           4  Female   23                  16                      77
4           5  Female   31                  17                      40

--- Missing Value Check ---
CustomerID              0
Genre                   0
Age                     0
Annual_Income_k$        0
Spending_Score_1-100    0
dtype: int64

--- Final Cluster Centroids & Profiles ---
 Cluster  Annual_Income_k$  Spending_Score_1-100  Customer_Count
       0         55.296296             49.518519              81
       1         86.538462             82.128205              39
       2         25.727273             79.363636              22
       3         88.200000             17.114286              35
       4         26.304348             20.913043              23
```







Labs Sheet Questions

1. What is clustering? How does it differ from classification?

- **Clustering:** An unsupervised learning technique used to partition unlabeled data into natural groupings based on structural or geometric similarities.
- **Difference from Classification:**
    - *Clustering (Unsupervised):* Works without predefined target classes or ground-truth labels. The model discovers patterns purely from feature similarities.
    - *Classification (Supervised):* Trains on labeled input-output pairs to map features to predefined target categories.

2. Explain the working principle of the K-Means Clustering algorithm.

K-Means partitions data into K distinct clusters by minimizing the Within-Cluster Sum of Squares (Inertia):

1. **Initialization:** Select K initial points as cluster centroids (often using K-means++ to space them apart).
2. **Assignment:** Assign each sample to its nearest centroid based on Euclidean distance.
3. **Update:** Recalculate each centroid as the arithmetic mean of all data points assigned to that cluster.
4. **Convergence:** Repeat the assignment and update steps iteratively until centroids remain stationary or a maximum iteration limit is reached.

3. Why is feature scaling important before applying K-Means clustering?

K-Means computes pairwise distances (Euclidean distance) across feature dimensions. If features have different scales (e.g., Annual Income in thousands versus Spending Score scaled from 1 to 100), the feature with larger numerical values will dominate the distance metric. Standardization ensures all features contribute equally to the distance calculation.

4. What is the Elbow Method, and how does it help determine the optimal number of clusters?

The Elbow Method plots the Within-Cluster Sum of Squares (WCSS / Inertia) against a range of cluster counts (K). As K increases, WCSS decreases monotonically. The "elbow" point represents the inflection where the rate of decrease drops sharply, indicating the optimal balance between cluster compactness and model complexity.

5. What is the Silhouette Score? How is it used to evaluate clustering performance?

The Silhouette Score measures how well-separated and cohesive clusters are. For an individual sample i:

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

- $a(i)$: Mean intra-cluster distance to all other points in the same cluster.
- $b(i)$: Mean nearest-cluster distance to points in the closest neighboring cluster.

Values range from -1 to $+1$:

- **Close to $+1$:** Well-separated, dense clusters.
- **Around 0:** Overlapping cluster boundaries.
- **Negative:** Samples may be assigned to the wrong cluster.

6. Why is K-Means considered an unsupervised machine learning algorithm?

It operates strictly on unlabeled input features without target variables or ground-truth class assignments. The algorithm derives groupings autonomously based on feature geometry and proximity rather than learning from labeled examples.

7. Mention any four real-world applications of customer segmentation.

- **Targeted Marketing Campaigns:** Delivering custom promotions tailored to specific spending tiers.
- **Customer Churn Prevention:** Identifying declining activity patterns in at-risk segments to deploy retention offers.
- **Personalized Product Recommendations:** Tailoring catalog feeds and bundle offerings based on cluster purchasing habits.
- **Dynamic Pricing & Credit Limits:** Assigning credit card limits, financing options, or promotional pricing based on spending behavior and income levels.

8. What are the limitations of the K-Means algorithm?

- **Fixed K Requirement:** Requires defining the number of clusters beforehand.
- **Spherical Cluster Bias:** Assumes clusters are spherical and isotropic; fails to detect complex, non-linear, or elongated cluster shapes.
- **Sensitivity to Outliers:** Because cluster centers are calculated as means, extreme values distort centroid coordinates.
- **Sensitivity to Initialization:** Random centroid placement can lead to convergence at suboptimal local minima.

9. How can businesses use customer segmentation to improve marketing and customer retention?

- **Targeted Campaigns:** By identifying distinct groups (e.g., high-income/high-spending vs. budget spenders), companies avoid generic ads and run high-conversion campaigns tailored to each group's budget.
- **VIP Retention:** Dedicated loyalty perks, dedicated managers, and early access are allocated to top-tier spenders to maximize lifetime value.
- **Re-engagement:** Price-sensitive or low-engagement customers receive discount codes or flexible payment models (e.g., BNPL) before churning.

10. Compare K-Means Clustering with Hierarchical Clustering based on their working principles and applications.

Aspect	**K-Means Clustering PDF**	**Hierarchical Clustering (Agglomerative) PDF**
Working Principle	Partitions data by iteratively updating K centroids using Euclidean distance.	Builds a bottom-up tree by sequentially merging the closest pairs of clusters.
Cluster Count (K)	Must be chosen prior to running the model.	Does not require an initial K; the tree can be sliced at any height.
Scalability	$\mathcal{O}(n \cdot K \cdot I \cdot d)$; scalable to large datasets.	$\mathcal{O}(n^2 \log n)$ to $\mathcal{O}(n^3)$; computationally prohibitive for large datasets.
Output Representation	Discrete cluster assignments with centroid coordinates.	A dendrogram showing nested sub-clusters and hierarchical structure.
Common Uses	Customer segmentation, document clustering, and large-scale tabular analysis.	Gene expression analysis, taxonomic classifications, and exploratory studies on small datasets.

Experiment No. 7: Customer Segmentation using K-Means Clustering

Aim

To implement the K-Means Clustering algorithm on a real-world dataset for customer segmentation and analyze customer groups based on purchasing behavior and demographic characteristics.

Questions & Answers

Q1. What is clustering? How does it differ from classification?

Answer:

• **Clustering:** An unsupervised learning task that partitions unlabeled data points into distinct groups (clusters) based on intrinsic pattern similarities without predefined ground-truth labels.

• **Classification:** A supervised learning task that trains models on labeled historical datasets to map features to target category classes.

Aspect	Clustering	Classification
Learning Type	Unsupervised Machine Learning	Supervised Machine Learning
Data Labeling	Unlabeled data	Pre-labeled target classes
Goal	Group data points by natural affinity	Predict predefined class categories

Q2. Explain the working principle of the K-Means Clustering algorithm.

Answer:

K-Means partitions N observations into K distinct clusters through an iterative optimization procedure:

1. **Initialization:** Select K initial centroid coordinates randomly across feature space.

2. **Assignment Phase:** Calculate Euclidean distance between every data point and each centroid; assign points to their closest centroid.

3. **Update Phase:** Recalculate cluster centroids as the arithmetic mean of all data points currently belonging to that cluster.

4. **Convergence:** Repeat the assignment and update steps until centroid movements fall below a predefined tolerance threshold or cluster memberships stabilize.

Q3. Why is feature scaling important before applying K-Means clustering?

Answer:

K-Means utilizes distance calculations (typically **Euclidean distance**) to compute similarity between observations.

• Unscaled features measured on larger numeric scales (e.g., Annual Income: \15,000 - \140,000) will mathematically dominate distance equations compared to features on smaller numeric scales (e.g., Age: 18 - 70).

• Feature scaling (via **StandardScaler** or **MinMaxScaler**) transforms features to equivalent scales, ensuring equal feature weighting.

Q4. What is the Elbow Method, and how does it help determine the optimal number of clusters?

Answer:

• **Elbow Method:** A heuristic optimization tool used to locate the ideal number of clusters (K).

• **Mechanism:** Plots the **Within-Cluster Sum of Squares (WCSS)** / Inertia against a range of sequential K values.

• **Selection Criterion:** As K increases, WCSS naturally drops. The optimal K value corresponds to the visual "elbow point"—the point after which diminishing returns occur in WCSS reduction.

Q5. What is the Silhouette Score? How is it used to evaluate clustering performance?

Answer:

The **Silhouette Score** measures how similar an object is to its own cluster compared to neighboring clusters.

• **Formula:**

S(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}

Where:

• a(i) = Mean intra-cluster distance for point i.

• b(i) = Mean nearest-cluster distance for point i.

• **Range Interpretation:**

• Score range: **-1 to +1**.

• Near +1: Points are tightly packed within their cluster and far from neighboring clusters (Optimal).

• Near 0: Overlapping clusters.

• Negative value: Incorrect cluster assignments.

Q6. Why is K-Means considered an unsupervised machine learning algorithm?

Answer:

K-Means operates entirely on unlabeled training datasets (X). It does not rely on target response variables (y) or predefined guidance labels to group observations. Instead, it uncovers natural structure and hidden groupings based purely on statistical distances.

Q7. Mention any four real-world applications of customer segmentation.

Answer:

1. **Targeted Marketing Campaigns:** Tailoring promotional emails and advertisements to specific demographic clusters.

2. **Personalized Product Recommendations:** Recommending items based on historical purchase patterns of similar customer groups.

3. **Dynamic Pricing Strategy:** Offering targeted discounts to price-sensitive customer segments while providing premium tiers to high-value cohorts.

4. **Customer Churn Prevention:** Identifying declining engagement patterns in specific segments to trigger retention campaigns.

Q8. What are the limitations of the K-Means algorithm?

Answer:

1. **Sensitivity to Initial Centroids:** Random initial centroid selection can yield sub-optimal local minima solutions (mitigated by `K-Means++`).

2. **Predefined K Value:** Requires manual definition of cluster counts before running execution loops.

3. **Outlier Sensitivity:** Outlier data points pull and distort average centroid positions.

4. **Spherical Cluster Assumption:** Struggles with complex non-convex, non-spherical spatial cluster geometries.

Q9. How can businesses use customer segmentation to improve marketing and customer retention?

Answer:

• **Tailored Messaging:** Replaces generic mass broadcasts with cohort-specific advertising copy that addresses unique buyer preferences.

• **Budget Optimization:** Allocates high marketing budgets toward high-value, high-margin customer cohorts.

• **Proactive Loyalty Programs:** Identifies vulnerable high-spending clusters nearing inactivity and offers targeted incentives to retain them.

Q10. Compare K-Means Clustering with Hierarchical Clustering based on their working principles and applications.

Answer:

Feature	K-Means Clustering	Hierarchical Clustering
Working Principle	Iterative centroid assignment based on Euclidean distances	Builds a nested tree structure (Dendrogram) via Agglomerative or Divisive steps
Speed & Scalability	Fast O(N \cdot K \cdot I), scales well to massive datasets	Computationally heavy O(N^3), restricted to smaller/medium datasets
Cluster Shape	Assumes spherical, convex cluster shapes	Capable of discovering arbitrary, non-spherical cluster forms
Pre-specifying K	Requires defining K in advance	Does not require pre-specifying K; dendrogram can be cut at desired height